



USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***Fourth Semester B.Tech (Minor in IOT) Degree Examinations**

May 2025

EC2006 -1 – INTRODUCTION TO IoT PROTOCOLS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which components are essential for building an IoT system?
A) Software, hardware, and firmware B) Routers, switches, and modems
C) Sensors, actuators, and servers D) Batteries, transistors, and capacitors
2. What is the purpose of actuators in an IoT system?
A) To sense environmental data B) To process data collected by sensors
C) To execute physical actions based on received instructions D) To establish communication with other devices
3. Which of the following is an example of IoT application in healthcare?
A) Smart agriculture B) Remote patient monitoring
C) Intelligent transportation systems D) Smart grid management
4. What is the primary function of HTTP in web communication?
A) Secure data transmission B) Real-time bidirectional communication
C) File transfer D) Hypertext transfer
5. Which architectural view represents the logical structure of an IoT system?
A) Functional view B) Communication view
C) Physical view D) Security view
6. What is M2M communication in the context of IoT?
A) Mobile-to-Mobile communication, where smartphones exchange data using cellular networks B) Machine-to-Modem communication, where devices connect to a central modem for data transmission
C) Machine-to-Machine communication, where devices exchange data directly without human intervention D) Multimedia-to-Multimedia communication, where various media devices share content wirelessly
7. Which application layer protocol is primarily used for secure data transmission over the internet?
A) MQTT B) HTTPS
C) HTTP D) Telnet
8. Which communication technology is commonly used in IoT for short-range wireless communication?
A) 3G B) Satellite
C) WiMAX D) Bluetooth
9. In IoT, what does IP addressing primarily facilitate?
A) Physical device assembly B) Device power management
C) Device manufacturing process D) Device identification and communication over networks
10. What are the different levels of IoT deployment?
A) Edge, Fog, and Cloud B) Basic, Intermediate, and Advanced
C) Tier 1, Tier 2, and Tier 3 D) Intranet, Extranet, and Internet

11. Which communication protocol is commonly associated with Representational State Transfer (REST)?
 A) MQTT B) CoAP
 C) HTTP D) Telnet
12. What does SOAP stand for in the context of web communication protocols for connected devices?
 A) Simple Object Access Protocol B) Service-Oriented Application Protocol
 C) Secure Online Authentication Protocol D) Systematic Object Application Protocol
13. Which protocol is commonly used for real-time bidirectional communication between web clients and servers?
 A) Web Sockets B) HTTP
 C) Telnet D) FTP
14. Which of the following is NOT a message communication protocol for connected devices?
 A) MQTT B) CoAP
 C) HTTP D) Telnet
15. What is the primary purpose of using Web Sockets in web communication for connected devices?
 A) To establish a secure connection B) To enable real-time bidirectional communication
 C) To transfer large files efficiently D) To facilitate email communication
16. What is the main purpose of a gateway in a web connectivity setup for connected devices?
 A) To restrict internet access B) To facilitate communication between devices with different protocols
 C) To provide security patches D) To increase network speed
17. In the Cloud of Things paradigm, where are IoT services primarily hosted?
 A) On physical servers B) On IoT devices themselves
 C) In virtualized cloud environments D) On edge computing nodes
18. Which of the following best describes the role of physical servers in IoT?
 A) Physical servers host IoT platforms and middleware B) Physical servers are unnecessary in IoT due to cloud computing
 C) Physical servers only handle data storage in IoT D) Physical servers are responsible for device-to-device communication in IoT
19. What distinguishes the Web of Things (WoT) from the Internet of Things (IoT)?
 A) WoT focuses on physical devices, while IoT focuses on virtual devices. B) WoT emphasizes web standards for device interaction, while IoT does not.
 C) WoT operates solely in cloud environments, while IoT operates on physical servers D) WoT devices are connected through browsers, while IoT devices are not.
20. Which of the following is not a component of IoT building blocks?
 A) Sensors B) Actuators
 C) Routers D) Servers

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Why CoAP is preferred in IoT application layer? Illustrate with an example.	8	L*2	1	1
	b) Distinguish between IPv4 and IPv6 protocol along with its addressing scheme used in network layer.	8	L2	2	1
2.	a) Discuss Bluetooth and Zigbee protocol in detail.	8	L2	1	1
	b) Describe the role of information and communication infrastructure in IoT.	8	L2	2	1
3.	a) What are the building blocks required to transform a simple sensor to IoT based sensor node? Illustrate with an example.	8	L2	1	1
	b) What are different network layer protocols for an IoT applications? Explain any one protocol along with its features.	8	L2	2	1
Unit – II					
4.	a) Illustrate the publish/subscribe communication with an example.	8	L2	3	1
	b) Explain the functionalities of Big data and Big stream in IoT.	8	L2	4	1

- | | | | | | | |
|----|----|--|---|----|---|---|
| 5. | a) | Illustrate the steps involved in packet data transfer of TCP Protocol. | 8 | L2 | 3 | 1 |
| | b) | What is IoT gateway? How IoT gateway can be implemented? | 8 | L2 | 4 | 1 |
| 6. | a) | What is fog computing? How an IoT platform can be enabled with both fog and cloud computing? | 8 | L2 | 3 | 1 |
| | b) | List the methods used in service discover process. Explain any two listed protocols. | 8 | L2 | 4 | 1 |

Unit – III

- | | | | | | | |
|----|----|--|---|----|---|---|
| 7. | a) | What are the features required for IoT Hardware? Discuss Smart object hardware, with radio network interface and hardware, with wired communication interface. | 8 | L2 | 5 | 1 |
| | b) | How Raspberry Pi is configured to use for IoT application? Explain with an example. | 8 | L2 | 5 | 1 |
| 8. | a) | Write short notes on
(i) Arduino
(ii) Zolertia Z1 | 8 | L2 | 5 | 1 |
| | b) | Write the table representing data size and code size for RFC7228 classes of constrained devices. Discuss the different classes briefly. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
